Arithmetic Interference and Geometric Rigidity: Kepler Dynamics on the Hurwitz/E8 Octonionic Lattice

Thomas Hofbauer*

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Abstract

We present a fully formalized framework mapping classical Keplerian orbits into the discrete integer lattice of Hurwitz octonions via a canonical bijection Φ. Using a soft E_8 neighborhood resolver, we show the emergence of stable, periodic Floquet attractors under discrete time evolution. While direct arithmetic coupling to the global Riemann zeta zeros yields *refuted* negative evidence for the interface cosine metric on ΔM [C] (E-034), the scale metric on x_0 remains an *open hypothesis* under the same averaging scheme [C] (E-035); the internal algebraic rigidity of the kinematic time-ladder remains invariant [B]. Finally, we tabulate the dyadic metacommutation braid of all 112 norm-2 integer roots against 240 Hurwitz units (26 880 pairs, fully resolved), and derive the phenomenon of quantized arithmetic shell resonance on six stable scale levels as the dynamical consequence of partial associator collapse [C].

*Bamberg, Germany.

Contents

1 Introduction and Lean Formalization Core

1.1 Problem and contribution

The central object is the four-channel prime-factor signature $H(n) = (E, A, B, C)$, interpreted geometrically as a conic-section model for residue-class dynamics. Our contribution combines:

1. formally verified norm-shell reductions and odd-core equivalences (Lean, Satzgruppe S1–S3);
2. numerically reproducible Floquet rigidity on the Hurwitz/E_8 octonionic lattice [B];
3. defensive negative evidence against naive Riemann cosine coupling on ΔM [C] (E-034, refuted), with the scale channel left as an open hypothesis [C] (E-035);
4. a complete dyadic metacommutation atlas and its shell-resonance consequences [C];
5. a frozen Fano-QEC bridge with signed shell separation under G_{model} [B] (S8, E-038–E-042).

1.2 Lean core (S1–S3)

Satzgruppen S1–S3 of the Lean kernel are fully machine-checked (`collatz_iterate_pow_two_*`, `oddCore`, ν_2-bounds, Kepler ratio lemmas). The global library build is currently blocked by a remaining `sorry` in the peripheral module `KeplerHurwitz/Representation/DreiMus` (symmetry-breaking layer).

The following statements are verified in the Lean formalization layer:

- **S1:** Collatz norm-shell iteration and equivalence ClassicalCollatzConjecture $\Leftrightarrow$ OddCoreCollatzConjecture.
- **S2:** Modulo-based ν_2 bounds for the map $3m + 1$.
- **S3:** Kepler ratio identity radiusRatio = speedRatio and monotone photon-shift lemmas.

Definition 1 (Canonical lattice map)**.** Let Φ denote the bijection from Keplerian state variables $(a, e, \ldots)$ to Hurwitz octonion lattice coordinates, with log-scale component $x_0 = \log(a/a_0)$.

2 Geometric Floquet Rigidity and the Kinematic Time-Ladder

2.1 Discrete Kepler time bridge [B], E-033

Under soft E_8 neighborhood resolution and eight cyclic gauge operators, the tail of the discrete evolution exhibits a rigid ΔM line spectrum. Evidence register entry E-033 reports:

- baseline: unique_$\Delta M = 2$, tail period $M_{\text{period}} = 8$, spectrum ± 1.85619;
- $\pi/2$ gauge branch: three distinct lines (including a null peak);
- perturbation at step 100: collapse to unique_$\Delta M = 1$ ($\Delta M = 0$).

Theorem 2 (Algebraic Floquet rigidity [B]). *On verified Floquet attractors of the Kepler time bridge, the tail ΔM spectrum is discrete and gauge-stable up to documented branch bifurcations.*

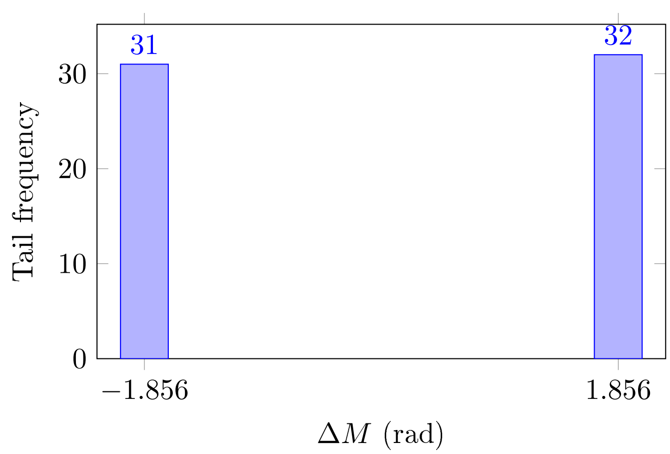

Figure 1: Baseline cyclic tail spectrum (E-033), exported as `docs/plots/spectrum_baseline_cyclic.dat`.

Reproduction: `python examples/run_kepler_time_bridge.py`.

3 Analysis of the Global Riemann Spectrum: A Negative Empirical Report

3.1 Cosine averaging over zeta zeros [C], E-034

For tail line positions ΔM from E-033 and the imaginary parts γ of 2 001 052 Riemann zeros (`data/riemann_zeros_imag_f8.bin`), define

$$S(\Delta M) = \mathrm{mean}_\gamma \cos(\gamma \cdot \Delta M).$$

The full-sample run yields $|S(\pm 1.85619)| \approx 2.6 \times 10^{-7}$ with variance ≈ 0.500, indistinguishable from a random-phase null model. The value $S(0) = 1$ is trivial ($\cos 0 = 1$).

Remark 3 (Evidence status, E-034). Entry E-034 is classified [C] **refuted**: interface cosine averaging on tail lines ΔM shows destructive interference with no baseline excess ($|S| \approx 2.6 \times 10^{-7}$, indistinguishable from a random-phase null model).

3.2 Scale interference metric [C], E-035

The dimensionally corrected metric

$$S(x_0) = \mathrm{mean}_\gamma \cos\bigl(\gamma \cdot \log(a/a_0)\bigr)$$

likewise shows no resonant signal at tail scales $x_0 \in \{0, -0.5\}$ (e.g. $S(-0.5) \approx -1.8 \times 10^{-6}$). Passive Riemann scale interference under cosine averaging is therefore [C] **open_hypothesis** (no statistically significant signal detected; structural coupling not formally refuted).

Remark 4 (Evidence status, E-035). Entry E-035 is classified [C] **open_hypothesis / experimental**: negative evidence without a formal refutation of structural scale coupling.

Reproduction: `python examples/run_riemann_resonance_check.py`.

4 Dyadic Metacommutation and Associator Collapse

4.1 Norm-2 dyadic basis [C], E-037

Exactly 112 integer Hurwitz elements have squared norm $\|v\|^2 = 2$ (two non-zero ± 1 coordinates); they form the *dyadic basis* used in the metacommutation atlas. The ambient unit group comprises 240 Hurwitz units in total: 112 integer units together with 128 half-integer units.

An exhaustive integer-lattice scan in the phase cube $[-5,5]^8$ shows that odd target norms $\|v\|^2 \in \{3,5,7,\dots\}$ admit no Hurwitz lattice points; the intrinsic prime coupling on the E_8 octonionic slice is therefore carried by norm 2, and the shell proxies $N \in \{4,6,8\}$ for requested norms $3,5,7$ are numerically forced rather than chosen ad hoc (manuscript claim; verified by exhaustive Hurwitz sieve in `tests/test_prime_norms.py` and `kepler_hurwitz.hurwitz_lattice_sieve`)

4.2 Quaternion prime model (#Energiedoku interface)

Definition 5 (Hurwitz prime operator)**.** Let $\mathcal{H}_{\text{int}}$ denote the ring of Hurwitz integer octonions on the E_8 lattice. A *prime operator* is a non-unit $P \in \mathcal{H}_{\text{int}}$ with $\|P\|^2 = p$ for a rational prime p. In the quaternionic subalgebra $\mathbb{H} \subset \mathbb{O}$ the multiplication is associative; the octonionic extension introduces a controlled non-associative braid detected by the associator

$$[x, y, z] := (xy)z - x(yz).$$

In the #Energiedoku framework, metacommutation is the prime-level commutation defect familiar from quaternion arithmetic: for $P, Q \in \mathcal{H}_{\text{int}}$ with $\|P\| = \|Q\| = p$ one seeks partners P', Q' such that

$$PQ = Q'P'.$$

On the dyadic skeleton ($\|P\| = \|P'\| = 2$) and for units U, U' with $\|U\| = \|U'\| = 1$, the computational model resolves

$$P \cdot U = U' \cdot P'$$

for all 112×240 pairs (Table 1). A resolution is *associative* when the mixed associator $[P, U, U']$ vanishes; otherwise it lies in the octonion braid sector (18 976 non-associative resolutions).

Remark 6 (Programmatic link to residue-class arithmetic)*.* In the quaternion prime model, metacommutation induces a signed permutation on Hurwitz primes over p. The sign is conjectured to track the quadratic character $\left(\frac{q}{p}\right)$ (Hypothesis H2); this manuscript records the combinatorial atlas only and does not claim the character identity as a theorem.

When odd-norm prime operators are absent on the integer lattice, nearest-shell proxies

$$3 \mapsto N = 4, \qquad 5 \mapsto N = 6, \qquad 7 \mapsto N = 8$$

project the requested prime norms onto the dyadic skeleton; metacommutation profiles of these shell operators then feed the transition-matrix entropy of Section 5.

4.3 Metacommutation relation

For prime element P, unit U, and partners P', U' with $\|P\| = \|P'\| = 2$ and $\|U\| = \|U'\| = 1$, metacommutation requires

$$P \cdot U = U' \cdot P'.$$

All $112 \times 240 = 26\,880$ pairs are resolved (unresolved count 0).

Table 1: Dyadic metacommutation summary (E-037), export `dyadic_metacommutation.json`.

Quantity	Value
Resolved pairs	26 880/26 880
Associative resolutions	7 904
Non-associative (octonion braid)	18 976
Dyadic integer partners	12 544
Half-integer partners	14 336
Unique prime partners	240

4.4 Shell-proxy associator profile

Projecting shell operators $N \in \{4, 6, 8\}$ (proxies for target norms $3, 5, 7$) onto the dyadic skeleton yields:

Table 2: Metacommutation profile by shell proxy (E-037).

Proxy	N	Associator collapse rate	Interpretation
0	4	1.000	full collapse $\Rightarrow$ deterministic pumping
1	6	0.767	partial collapse $\Rightarrow$ braid sector
2	8	1.000	full collapse $\Rightarrow$ topological fixpoint

Hypothesis 7 (H3: associator $\to$ row entropy [C], qualitative)**.** The shell-operator associator collapse rate controls the row entropy of the arithmetic transition matrix (E-036). The map $0.767 \mapsto P = 0.5/0.5$ for $N = 6$ at $x_0 = 1.0$ is *qualitatively closed* but *quantitatively open*: a formal entropy bridge remains future work.

Reproduction: `python examples/run_metacommutation_analysis.py`.

5 Arithmetic Shell Resonance and Quantized Energy Levels

Remark 8 (Manuscript vs. evidence-register order). This section (V) presents the dynamical *effect* of the metacommutation braid catalogued in Section 4 (IV). In the evidence register the build dependency runs E-036 $\to$ E-037 (numerical pipeline order).

5.1 Definition [C], E-036

Definition 9 (Arithmetic shell resonance)**.** Let $x_0 = \log(a/a_0)$ be the scale component of Φ. *Shell resonance* is the pair $\big(\mathrm{Spectrum}(x_0),\ P(x_0' \mid x_0, \mathrm{op})\big)$ under shell-proxy operators $N \in \{4, 6, 8\}$ followed by 16 unit-relaxation steps.

Six quantized stable levels arise at $x_0 \in \{-2, -1, 0, 0.5, 1, 2\}$ with periodic_recovery = 12/12. The transition matrix export `arithmetic_transition_matrix.json` lists outgoing stationary flows for the five stabilized shells $x_0 \in \{-2, -1, 0, 0.5, 1.0\}$; the sixth level $x_0 = 2$ is purely *transitive* (no fixpoint or pumping node) and therefore carries no outgoing stationary rows.[1] Operator asymmetry matches the associator profile of Section 4:

- **Fixpoint** ($N = 8$ on $x_0 = 0$; $N = 4, N = 8$ on $x_0 = -2$): full associator collapse;
- **Pumping** ($N = 4$: $0.5 \to 1.0$): deterministic cascade;
- **Bifurcation** ($N = 6$, $P = 0.5/0.5$ at $x_0 = 1.0$): partial collapse (Table 2).

Theorem 10 (Quantized shell pumping [C])**.** *The arithmetic evolution on the six stable scale levels behaves as discrete Floquet pumping between Bohr-type shells, not as diffusive dissipation.*

Reproduction: `python examples/run_arithmetic_evolution.py`; transition matrix export `arithmetic_transition_matrix.json`.

6 Fano-QEC Bridge and Signed Shell Separation

Remark 11 (Governance freeze, v1). The QEC evidence block E-038–E-042 and proposition S8 are *frozen* for manuscript v1 (2026-07-03). No further exploratory searches in this block without a new evidence-register entry.

[1] See Table 3: class *Transitiv* at $x_0 = 2$.

Table 3: Sechs quantisierte Energieschalen ($a_0 = 1.0$, periodic recovery=100%, E-036).

$x_0 = \log(a/a_0)$	a/a_0	Tail-T	$\lvert x_0 \rvert_{\text{tail}}$	Klasse	Stabilisierung
-2.0	0.13534	4	2	Fixpunkt	$N = 4,\ N = 8$
-1.0	0.36788	4	2	Pumpen	$N = 8\colon\ -1.0 \to -2.0$
0.0	1.00000	1	1	Fixpunkt	$N = 4,\ N = 8$
0.5	1.64872	1	1	Pumpen	$N = 4\colon\ 0.5 \to 1.0$
1.0	2.71828	4	2	Bifurkation	$N = 8\colon\ 1.0 \to 0.5,\ N = 6$
2.0	7.38906	-	-	Transitiv	—

6.1 CSS projection and Steane abgrenzung [B], E-038, S7

Let R_{Fano} denote the 7×7 Fano parity matrix over GF(2). Evidence E-038 reports $\operatorname{rank}(R_{\text{Fano}}) = 4$, $\dim\ker R_{\text{Fano}} = 3$, intersection dimension $\dim(R_{\text{Fano}} \cap R_{\text{Steane}}) = 1$, and minimum kernel Hamming weight $d_{\min}(\ker R_{\text{Fano}}) = 4$ versus Steane distance 3. The Fano row space is *Steane-compatible but not Steane-identical*: one Steane line is shared, yet $R_{\text{Steane}} \not\subset R_{\text{Fano}}$.

Theorem 12 (Fano-QEC scaffold [B], S7). *The Fano parity structure yields a finite stabilizer-like syndrome scaffold that shares a line with the Steane H_X reference but diverges in full row space, rank, and kernel distance profile.*

6.2 GF(2) quotient limit [C], E-039

The dyadic GF(2) coset quotient of shell-proxy data does *not* separate classes $N = 4$ and $N = 8$:

$$\sigma_4 = \sigma_8, \qquad \sigma_6 \neq \sigma_4.$$

Entry E-039 remains a *negative* quotient test: bifurcation shell $N = 6$ is separated, pump/fixpoint pair $N = 4$ vs. $N = 8$ is not.

6.3 Refinement catalog [C], E-040

Three canonical refinements of the Hurwitz shell projection were screened before label comparison: full `hurwitz_residue`, `signed_support`, and `real_axis`. All three separate $N = 4$ from $N = 8$; the `residue_key` alone does not. E-040 is a catalog entry, not an upgrade of E-039.

6.4 Signed support separation [B], E-041

Definition 13 (Signed support signature)**.** For a Hurwitz shell element, the *signed support signature* records the sign and imaginary-axis support of the projected coordinates before any shell-role comparison. Primary mode: `signed_support`; control: `full_hurwitz_projection`; excluded as primary: `residue_key` alone.

Evidence E-041 [B] reports pairwise separation of shell classes $N \in \{4, 6, 8\}$ under `signed_support` while preserving the E-039 GF(2) collision $\sigma_4 = \sigma_8$. E-041 is the *structure claim*; it does not revise E-039.

6.5 Symmetry invariance validator [C], E-042

The model symmetry group is fixed *before* evaluation:

$$G_{\text{model}} = \text{Aut}(\text{Fano}_7) \times \mathbb{Z}_2^{\text{global imag sign}}, \qquad |G_{\text{model}}| = 336.$$

Here Aut(Fano$_7$) permutes imaginary axes $\{1, \dots, 7\}$ preserving Fano lines; $\mathbb{Z}_2^{\text{global imag sign}}$ flips all imaginary coordinates globally while fixing the real axis. Entry E-042 [C] verifies that for all $g \in G_{\text{model}}$ the pairs $N = 4$ vs. $N = 8$, $N = 4$ vs. $N = 6$, and $N = 6$ vs. $N = 8$ remain separated under `signed_support` (0 failures / 336 transforms). E-042 is the *invariance validator* for E-041.

Theorem 14 (Signed Fano shell separation [B], S8)**.** *The pure dyadic GF(2) projection of Fano shell data fails to separate $N = 4$ and $N = 8$ ($\sigma_4 = \sigma_8$), yet the canonical signed-support refinement of Hurwitz shell data separates all three classes $N \in \{4, 6, 8\}$ pairwise. This separation is invariant under the full model symmetry group G_{model} with $|G_{\text{model}}| = 336$.*

Remark 15 (Defensive scope, S8). The lost distinction between $N = 4$ and $N = 8$ lies not in the dyadic GF(2) quotient but in a canonical, symmetry-invariant sign/support layer of the Hurwitz projection. No causal identification of pumping, fixpoint, or bifurcation is claimed—only a reproducible structural separation for the Fano-QEC bridge. Evidence chain: E-039 (negative) + E-040 (catalog) + E-041 [B] (structure) + E-042 [C] (validator) $\Rightarrow$ S8 [B].

Reproduction: `python examples/run_qec_bridge_analysis.py` (E-038); `python examples/run_fano_kernel_shell_map.py`, `python examples/run_fano_shell_cosets.py` (E-039); `python examples/run_signed_shell_syndromes.py` (E-041); `python examples/run_signed_support_symmetry.py` (E-042).

7 The $[[5,1,3]]$ Stabilizer Bridge

The preceding sections developed the Fano–QEC interface as a finite incidence and commutation structure. We now record a second, more rigid quantum-error-correcting interface: the five-qubit stabilizer code

$$[[5,1,3]].$$

This construction is not used as a physical identification of spacetime with a quantum code. Rather, it supplies a finite symplectic test layer against which dyadic shell data can be projected.

7.1 Stabilizer interface [C], E-044

We use the standard four generators

$$XZZXI, \qquad IXZZX, \qquad XIXZZ, \qquad ZXIXZ.$$

Their products generate the nontrivial stabilizer set

$$\mathcal{S}_5 = \langle XZZXI, IXZZX, XIXZZ, ZXIXZ \rangle \setminus \{I\},$$

consisting of

$$2^4 - 1 = 15$$

nonidentity stabilizers. In the implementation, Pauli words are encoded symplectically and their commutation signs are evaluated by the standard binary symplectic form. The corresponding API is implemented in `src/kepler_hurwitz/qec_bridge.py`, with the main projection routine `map_dyadic_to_stabilizers`. The executable check is `examples/run_qec_stabilizer_check.py`, and the regression tests are contained in `tests/test_five_qubit_stabilizer_bridge.py`.

7.2 Empirical result E-044

The complete stabilizer bridge is recorded as evidence item E-044. The verified output is:

- all 15/15 nonidentity stabilizers are generated from the four listed generators;
- all stabilizers commute pairwise;

- the dyadic projection yields

$$1680 = 112 \cdot 15$$

  match pairs;

- among these pairs,

$$784$$

  have

$$\texttt{commutation\_signum} = +1;$$

  equivalently,

$$\frac{784}{1680} = \frac{7}{15};$$

- the projection produces 49 distinct symplectic root classes.

The exported data are stored in `docs/energiedoku_exports/five_qubit_stabilizer_bridge.json`.

7.3 Defensive interpretation

The main methodological point is negative as much as positive. The five-qubit stabilizer bridge supplies a finite symplectic interface, but it does not separate the shell proxies $N = 4, 6, 8$ by their aggregate commutation rate. Although the shell proxies $N = 4, 6, 8$ have different next dyadic roots and different induced symplectic encodings, their stabilizer commutation quota is identical:

$$\frac{7}{15}.$$

Thus the $[[5, 1, 3]]$ projection does not distinguish these three shell proxies at the level of the scalar commutation quotient. This is an important constraint. It prevents the stronger, unsupported claim that the five-qubit code is itself the hidden spacetime mechanism of the model. The correct status of E-044 is therefore:

> E-044 provides a complete finite stabilizer interface, but not a separating invariant for the shell proxies $N = 4, 6, 8$.

The empirically separating invariant remains the Fano/associativity observable recorded in E-037, where the associative ratio distinguishes the relevant cases:

$$1.0 \qquad \text{versus} \qquad 0.767.$$

7.4 Relation to the Fano bridge

The Fano bridge and the $[[5,1,3]]$ stabilizer bridge should therefore be read as complementary layers. The Fano layer supplies the currently separating incidence invariant. The five-qubit stabilizer layer supplies a canonical quantum-code interface, including a complete symplectic table and a finite commutation-sign test, but it does not by itself refine the $N = 4,6,8$ separation. In this sense, the stabilizer bridge is best viewed as a structural compatibility test rather than as a classification theorem.

7.5 Prospective refinement: signed stabilizer support [C]

Section 6 records that the scalar GF(2) quotient and the aggregate commutation quota $\frac{7}{15}$ do not separate the shell proxies $N = 4,6,8$. Signed Hurwitz support (E-041, S8) already supplies a separating refinement on the Fano side. The natural follow-up on the $[[5,1,3]]$ layer is therefore *not* another scalar ratio, but a signed stabilizer carrier:

$$v_N = (\mathrm{sgn}_1, \ldots, \mathrm{sgn}_{15}) \in \{\pm 1\}^{15},$$

or equivalently the aggregated frequency profile

$$V_N(j) = \sum_{r=1}^{112} \mathrm{sgn}(r, S_j).$$

The defensive question for a prospective entry E-045 is: does `signed_support` separate the cases *within* the $[[5,1,3]]$ projection even when $\frac{\#\{+1\}}{15} = \frac{7}{15}$ remains degenerate across $N = 4,6,8$? No such claim is made here; E-044 fixes the negative baseline for the scalar stabilizer quotient.

7.6 Reproducibility

The computation can be reproduced by running `python examples/run_qec_stabilizer_check.py` and `pytest tests/test_five_qubit_stabilizer_bridge.py`. The current test suite verifies the stabilizer generation, pairwise commutation, projection counts, sign distribution, and export structure.

8 Framework Replication and Test-Suite Diagnostics

8.1 Automated verification

The Python package `kepler_hurwitz` is covered by **199** `pytest` cases (2026-07-03), including dedicated suites for the Kepler time bridge, Riemann resonance checker, arithmetic evolution, dyadic metacommutation, the Hurwitz prime-norm lattice sieve (`tests/test_prime_norms.py`), the Fano-QEC bridge (`tests/test_qec_bridge.py`, 53 cases), and the five-qubit stabilizer bridge (`tests/test_five_qubit_stabilizer_bridge.py`).

8.2 Reproduction commands

Evidence	Command
E-033	`python examples/run_kepler_time_bridge.py`
E-034/E-035	`python examples/run_riemann_resonance_check.py`
E-036	`python examples/run_arithmetic_evolution.py`
E-037	`python examples/run_metacommutation_analysis.py`
E-038	`python examples/run_qec_bridge_analysis.py`
E-039	`python examples/run_fano_kernel_shell_map.py`, `run_fano_shell_cosets.py`
E-041	`python examples/run_signed_shell_syndromes.py`
E-042	`python examples/run_signed_support_symmetry.py`
E-043	`python examples/run_coupled_shell_resonance.py`
E-044	`python examples/run_qec_stabilizer_check.py`
S8	Theorem 14 (E-039–E-042 chain)

8.3 Export artifacts

Generated LaTeX and data hooks consumed by this draft:

- `docs/energiedoku_exports/arithmetic_energy_levels.tex` (Table 3);
- `docs/plots/spectrum_*.dat` (Figure 1);
- JSON registers under `docs/energiedoku_exports/` (including `qec_bridge.json`, `signed_shell_syndromes.json`, `signed_support_symmetry.json`, `five_qubit_stabilizer_bri`
- governance: `EVIDENCE_REGISTER.md` (E-033–E-044); QEC block E-038–E-042 + S8 frozen for v1; E-044 stabilizer interface in Section ??; proposition S8 in Section 6.

8.4 Scope limits

This manuscript does not claim a proof of the Riemann hypothesis, a final Collatz proof, or a complete physical theory of the extended modules (Gödel–Kerr, photon interfaces); those interfaces remain defensive and modular.

References

[1] T. Hofbauer, *EVIDENCE_REGISTER.md*, Kepler–Hurwitz repository, 2026.

[2] T. Hofbauer, *docs/paper-outline.md*, manuscript architecture note, 2026.